

Neutral Territory & Hybrid Decompositions for Parallel MD

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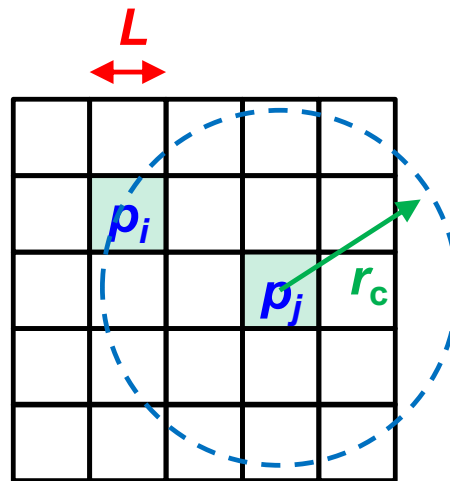
Who does what of MD?

D. E. Shaw, [A fast, scalable method for the parallel evaluation of distance-limited pairwise particle interactions](#), *J. Comput. Chem.* **26**, 1318 ('05)



Fine Granularity

Number of atoms per process (N/P) ~ 1
cf. Biomolecular simulations



spatial subsystem length (L) $\ll$ interaction cutoff (r_c)

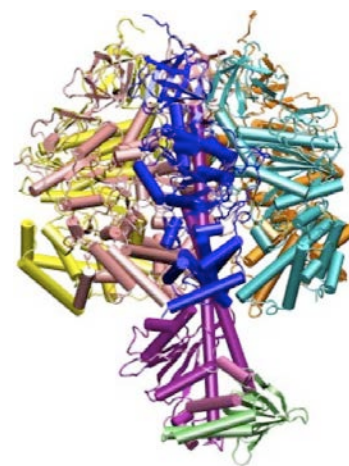
Fine-Grained Parallel MD

Pathways to a Protein Folding Intermediate Observed in a 1-Microsecond Simulation in Aqueous Solution

Yong Duan and Peter A. Kollman*

An implementation of classical molecular dynamics on parallel computers of increased efficiency has enabled a simulation of protein folding with explicit representation of water for 1 microsecond, about two orders of magnitude longer than the longest simulation of a protein in water reported to date. Starting with an unfolded state of villin headpiece subdomain, hydrophobic collapse and helix formation occur in an initial phase, followed by conformational readjustments. A marginally stable state, which has a lifetime of about 150 nanoseconds, a favorable solvation free energy, and shows significant resemblance to the native structure, is observed; two pathways to this state have been found.

Science **282**, 740 ('98)

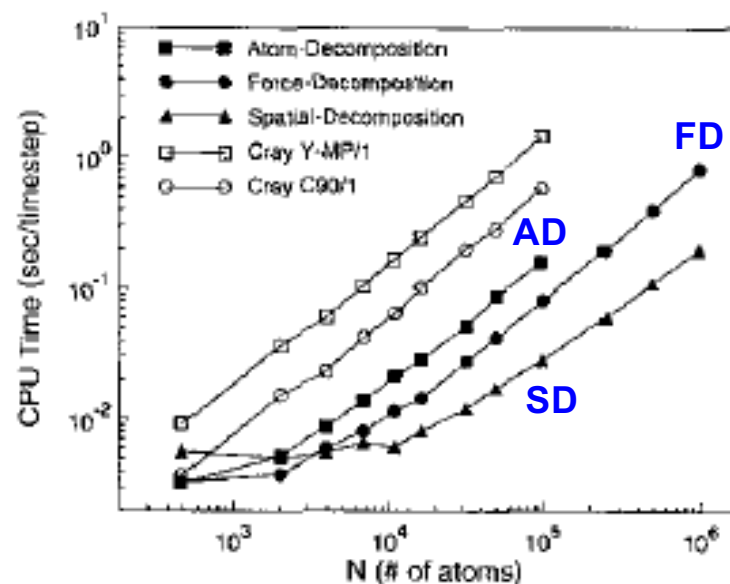
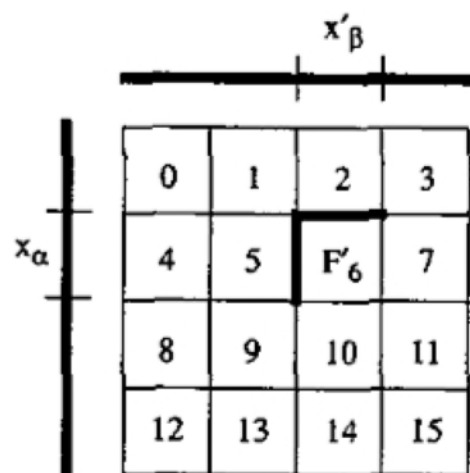


Processors		Time/step		Speedup		GFLOPS	
Total	Per Node	MPI	Elan	MPI	Elan	MPI	Elan
1	1	28.08 s	28.08 s	1	1	0.480	0.480
128	4	248.3 ms	234.6 ms	113	119	54	57
256	4	135.2 ms	121.9 ms	207	230	99	110
512	4	65.8 ms	63.8 ms	426	440	204	211
510	3	65.7 ms	63.0 ms	427	445	205	213
1024	4	41.9 ms	36.1 ms	670	778	322	373
1023	3	35.1 ms	33.9 ms	799	829	383	397
1536	4	35.4 ms	32.9 ms	792	854	380	410
1536	3	26.7 ms	24.7 ms	1050	1137	504	545
2048	4	31.8 ms	25.9 ms	883	1083	423	520
1800	3	25.8 ms	22.3 ms	1087	1261	521	605
2250	3	19.7 ms	18.4 ms	1425	1527	684	733
2400	4	32.4 ms	27.2 ms	866	1032	416	495
2800	4	32.3 ms	32.1 ms	869	873	417	419
3000	4	32.5 ms	28.8 ms	862	973	414	467

J.C. Phillips, G. Zheng, S. Kumar, & L.V. Kale,
in *Proc. of IEEE/ACM SC2002*

Table 1: NAMD performance on 327K atom ATPase benchmark system with and multiple timestep-ping with PME every four steps for Charm++ based on MPI and Elan.

Force Decomposition for Parallel MD



Runtime on 1,024-processor Intel Paragon

FIG. 5. The division of the permuted force matrix F' among 16 processors in the force-decomposition algorithm. Processor P_6 is assigned a sub-block F'_6 of size $N/\sqrt{P}$ by $N/\sqrt{P}$. To compute its matrix elements it must know the corresponding $N/\sqrt{P}$ -length pieces x_α and x'_β of the position vector x and permuted position vector x' .

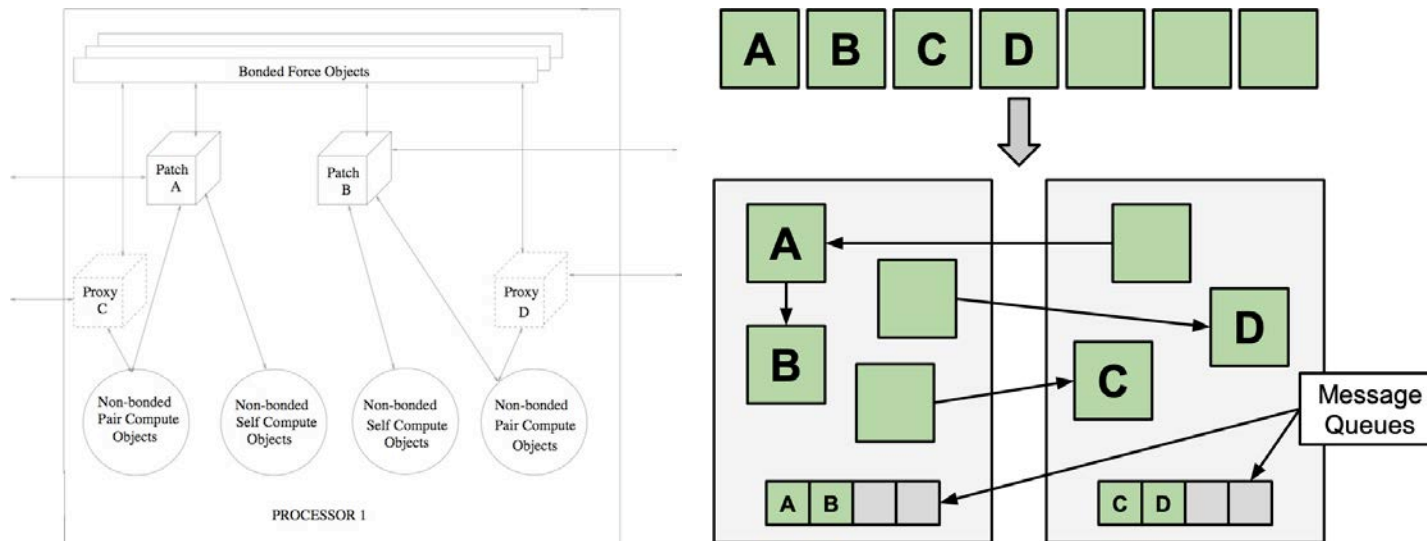
S. Plimpton, *J. Comput. Phys.* 117, 1 ('95)

<https://www.lammps.org/cite.html>



Hybrid Spatial+Force Decomposition

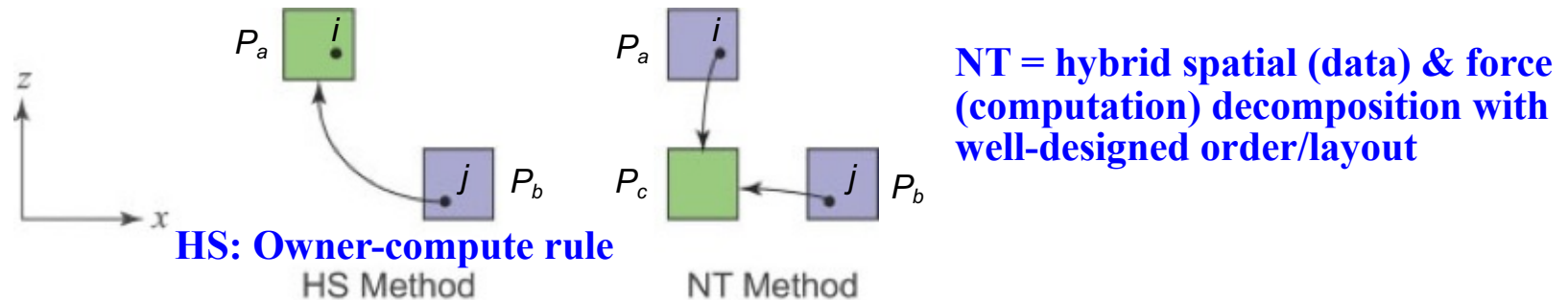
- Spatial decomposition of patches (localized spatial regions & atoms within)
- Inter-patch force computation objects assigned to any processor
- Message-driven object execution: computation-communication overlap



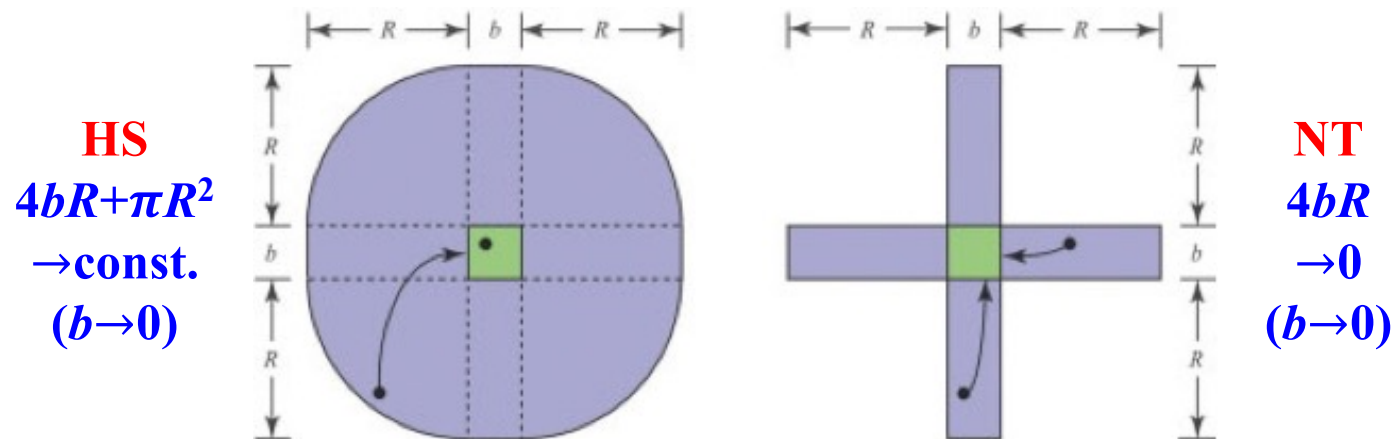
Kale *et al.*, *J. Comput. Phys.* **151**, 283 ('99); Phillips *et al.*, *SC02* (IEEE/ACM);
Acun *et al.*, *SC14* (IEEE/ACM), Phillips *et al.*, *J. Chem. Phys.* **153**, 044130 ('20)

Spatial (Half-Shell) vs. NT Decompositions

Locus of interaction — who does what (2-dimensional example)

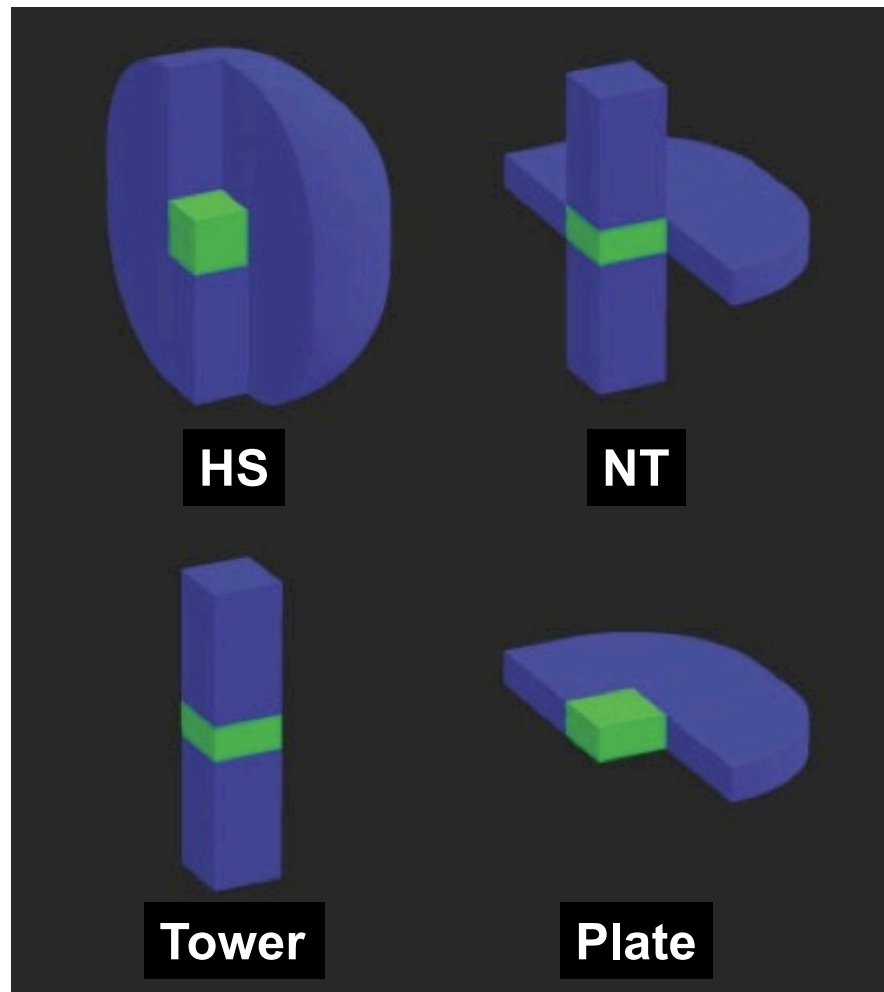


Import regions or communication volume (2-dimensional example)

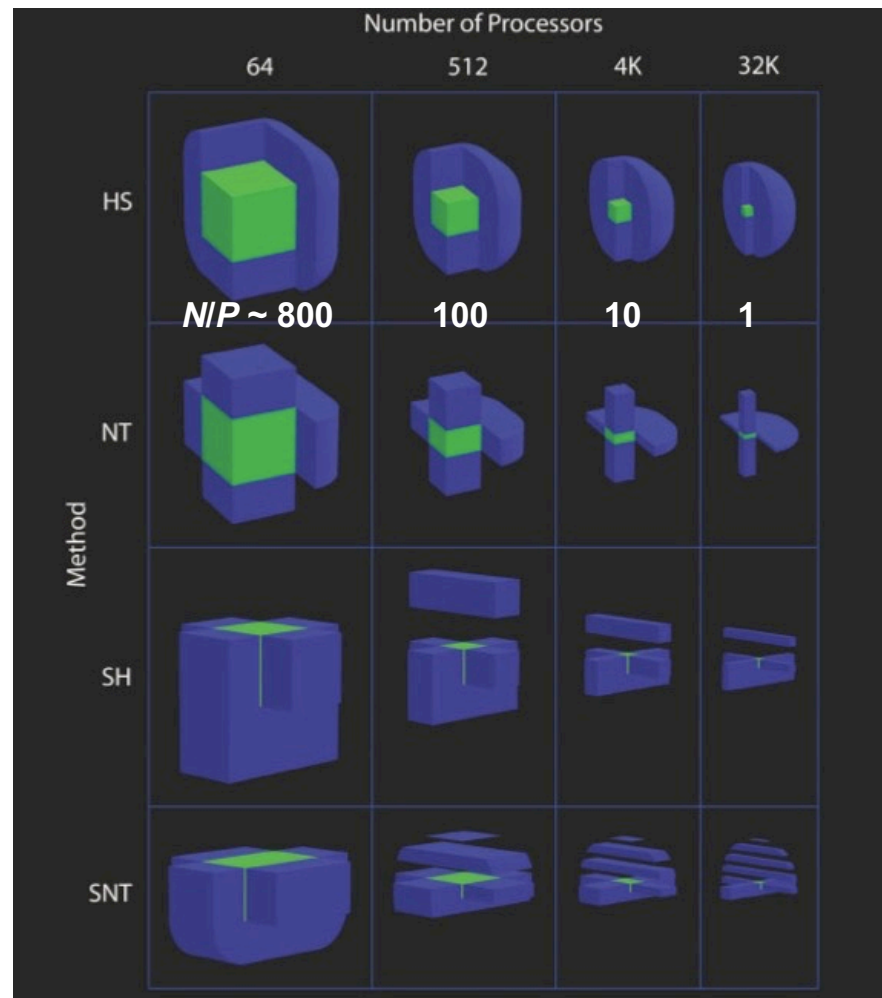


Import volume will be halved using Newton's 3rd law

3D Import Regions



Scaling of Import Regions



Marc Snir

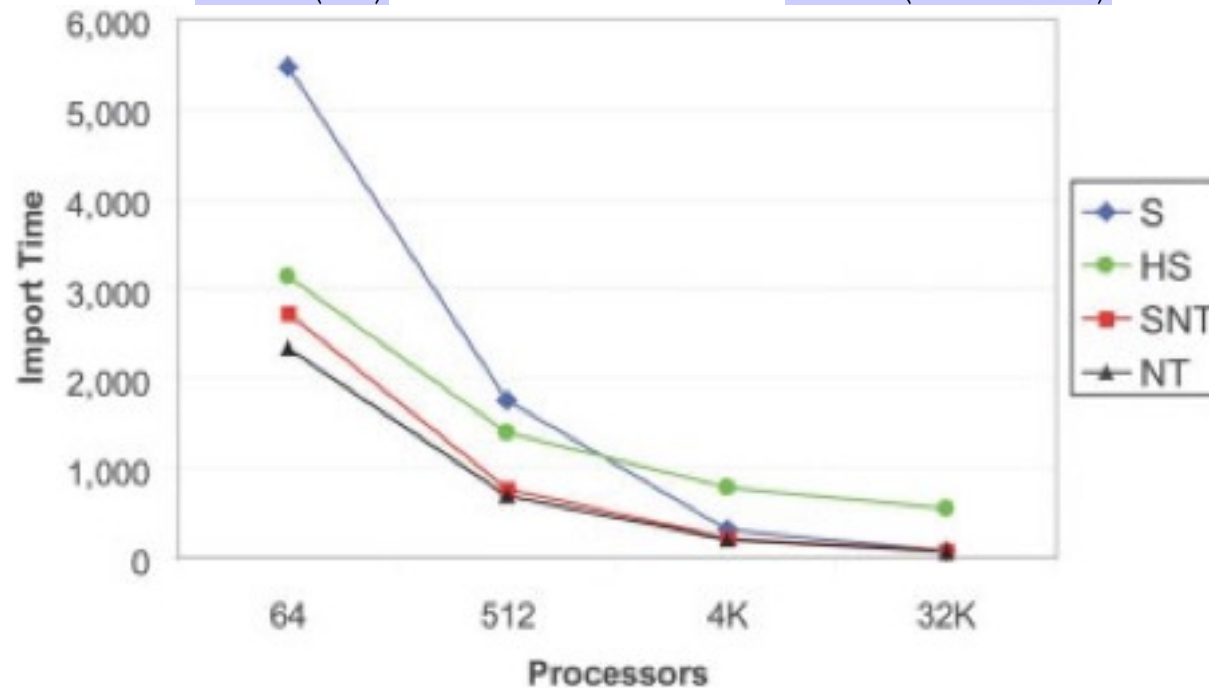
Scaling of the Volume of Import Regions

HS decomposition

$$V_i = O(R^3)$$

NT decomposition

$$V_i = O(R^{3/2} p^{-1/2})$$



Communication time

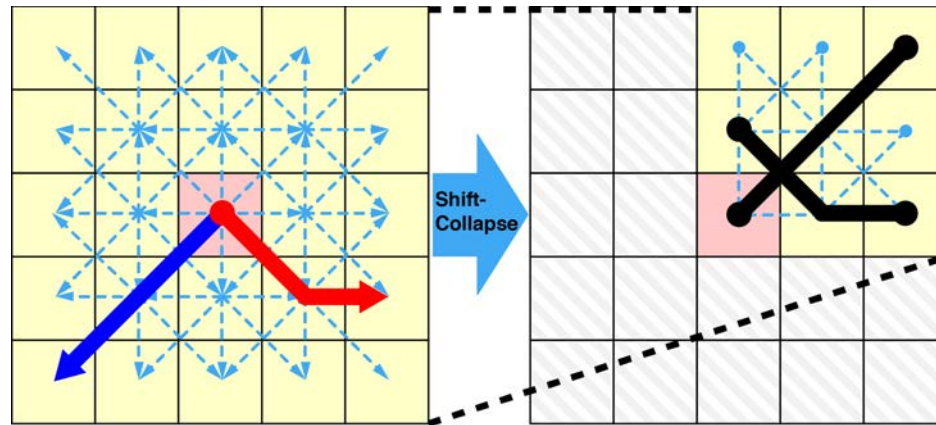
$$T_{\text{comm}} = t_{\text{latency}} \frac{\text{\# of messages}}{N_{\text{message}}} + \frac{1}{b_{\text{bandwidth}}} \frac{\text{volume (Bytes) of messages}}{V_{\text{message}}}$$

ns ~ many μ s

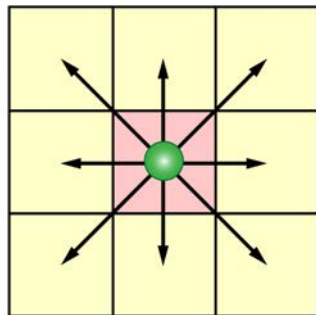
GB/s

Shift-Collapse (SC) Algorithm

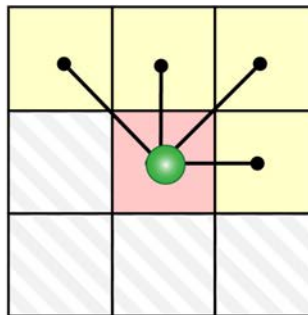
- Generalization of Shaw's eighth-cell method (non-owner-compute method on high-latency cluster) for pair computation to general dynamic range-limited n -tuples



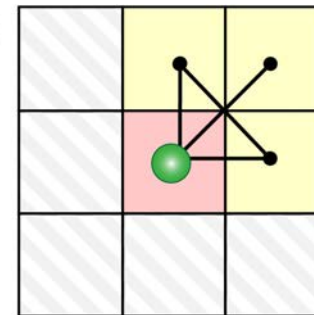
M. Kunaseth *et al.*, *IEEE/ACM Supercomputing (SC13)*



Full-shell (FS) method
[e.g. Rappaport, '88]



Half-shell (HS) method
[e.g. Rappaport, '88]

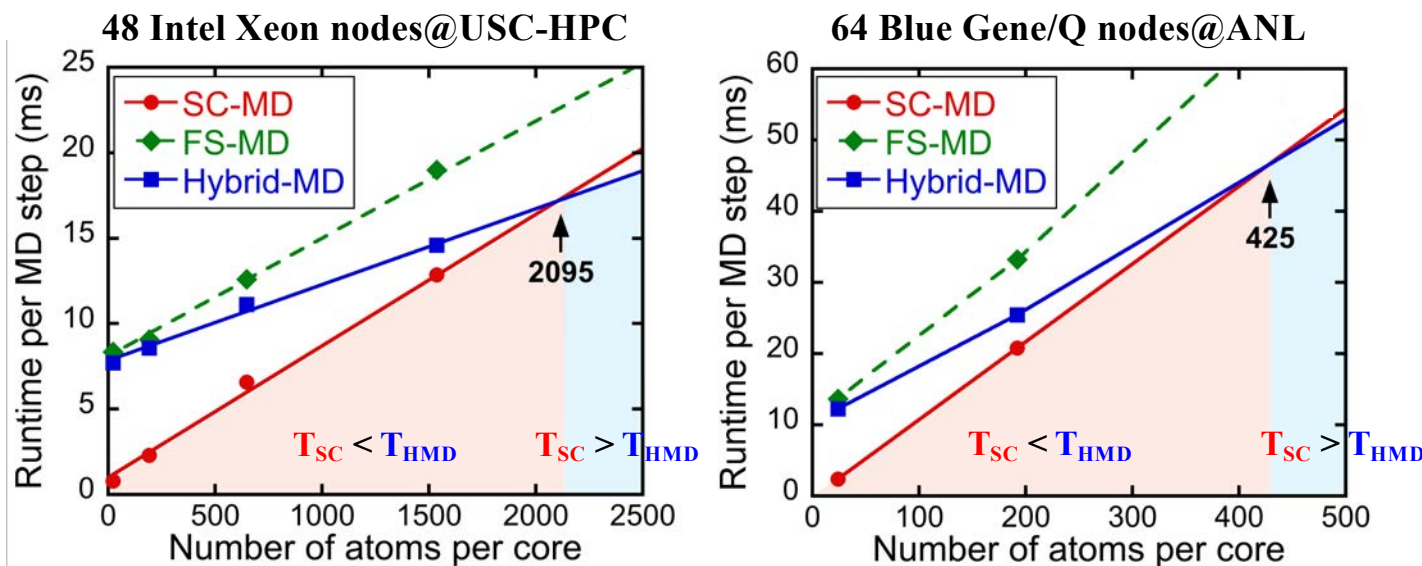


Eighth-shell (ES) method
[Bower *et al.*, '06]

Shift-Collapse (SC) Performance

Runtime comparison on 48 Intel-Xeon nodes and 64 Blue Gene/Q nodes

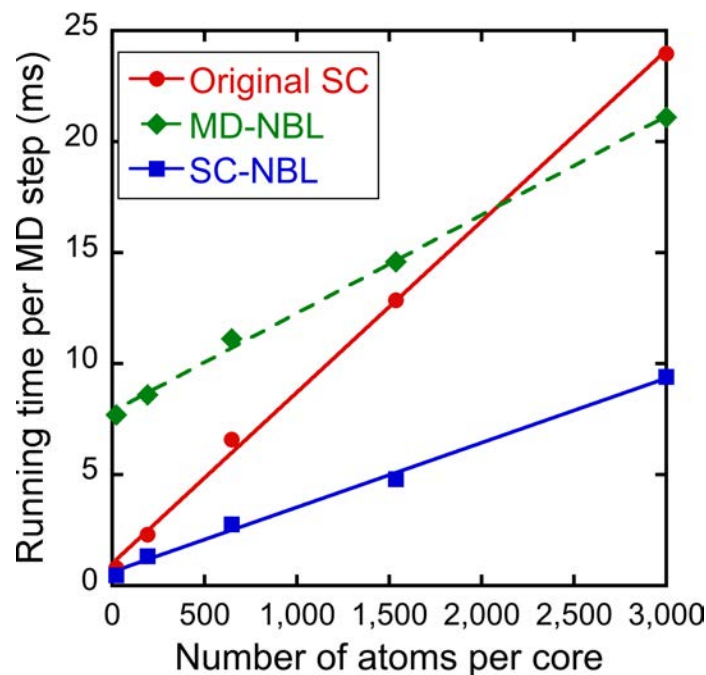
- SC-MD is always faster than FS-MD
- At the smallest grain, SC-MD is 9.7- and 5.1-fold speedups over the state-of-the-art hybrid linked-cell & neighbor list code
- Crossover of optimal algorithm from SC-MD to hybrid MD at larger granularity (*i.e.*, $N/P > 2,095$ on Intel Xeon and $N/P > 425$)



M. Kunaseth et al., [IEEE/ACM Supercomputing \(SC13\)](#)

Shift-Collapse on Neighbor List (SC-NBL)

- Apply shift-collapse operations to the hybrid linked-cell & neighbor list code (best of both)



[Shift/collapse on neighbor list \(SC-NBL\): fast evaluation of dynamic many-body potentials in molecular dynamics simulations](#), M. Kunaseth, S. Hannongbua, & A. Nakano, *Comput. Phys. Commun.* **235**, 88 (2019)

Challenge: Expose massive data parallelism for SC on graphics processing unit (GPU)